

Chitrarth Rai

chitrarthrai10@gmail.com | +91 7357084507 | linkedin.com/in/chitrarth-rai-38a40917b | github.com/Chitrarthrai

PROFILE

Graduate of IIIT Bhubaneswar in Electronics & Telecommunication. Full-stack developer specializing in the MERN ecosystem and Next.js, with expertise in building scalable HR tech, digital shelf analytics, and financial dashboards. Architected, analyzed, and contributed to over 40+ distinct software projects and microservices. Proficient in leveraging advanced algorithms and cloud functions (Azure) to deliver high-performance, data-driven applications.

TECHNICAL SKILLS

Languages/Frameworks	TypeScript, Python, C++, Kotlin, JavaScript (ES6+), React.js, Next.js, Node.js, Express.js, React Native
Databases/Infra	MongoDB, PostgreSQL, Firebase, REST APIs, JSON-RPC, Azure Blob Storage, Azure Functions
Tools/Practices	Git/GitHub, Android NDK/SDK, OpenCV, OCR, OpenAI API, Performance Optimization, System Design

EXPERIENCE

Neophyte AI Software Engineer	Feb 2025 – Present Navi Mumbai, India
Neo Disha (mobile): Designed a high-performance camera pipeline using <i>React Native</i> and a custom <i>Kotlin Native Module</i>. Optimized CPU-intensive image processing passes for 12MP sensor arrays, reducing processing latency by 40%. Constructed a precision coordinate mapping system featuring a gesture-driven <i>Reanimated</i> UI with sub-pixel accuracy, accelerating manual shelf adjustments. NeoQCR (web & Android): Modernized legacy Android builds (Gradle/NDK/SDK migration) to support the latest OS versions, ensuring compatibility with custom AI inference models. Created a unified dashboard for store-wise media management and KPI visualization, enhancing data accessibility. Disha (web): Built a shelf-analytics dashboard that ingests anomaly data from MongoDB and Azure Blob Storage; implemented real-time KPI tracking via interactive charts to improve operational visibility by 35%. HRMS platform: Architected a full-scale HR system featuring <i>Geolocation</i> and <i>Facial-recognition</i> based attendance; implemented complex <i>MongoDB Aggregation Pipelines</i> for high-speed payroll and leave balance rollups, cutting processing time by 50%. Interactive UI: Integrated <i>Three.js</i> for 3D product showcases and crafted a reusable React component library, standardizing analytics dashboards across multiple in-house initiatives and reducing frontend development time. - Participated in driving forward additional initiatives including NEOQA, NEOCompliance, and Pocket AI, providing robust technical solutions for cross-functional teams.	

PROJECTS

FinanceTask <i>React, TypeScript, Supabase, AI, React Native, Tailwind</i>	Code
- Architected a unified financial and productivity ecosystem that seamlessly synchronizes data between Web and Mobile platforms in real-time via Supabase WebSockets. - Constructed a comprehensive Reporting module offering P2P financial comparisons and automated monthly PDF summaries, boosting user financial visibility. - Programmed a full-featured task management system with Kanban boards and integrated 'Smart Notes' to automatically extract actionable tasks using AI. - Integrated Gemini AI to create a context-aware assistant delivering proactive financial insights and detecting budget anomalies with high accuracy. - Designed a high-performance KPI dashboard using Recharts to visualize real-time pocket money allocation and auto-adjust daily limits based on savings goals.	
FedEx Document Processing Pipeline <i>Python, OpenAI API, OCR, OpenCV</i>	Code
Authored an automated document analysis script using Python, OCR, and OpenCV to extract text, leveraging OpenAI to structure complex data and eliminate manual entry hours.	
Neo Disha (Native App) <i>React Native, Kotlin Native Module, Reanimated</i>	Code
- Designed a high-performance camera pipeline, optimizing CPU-intensive image processing passes for 12MP sensors. - Constructed a precision coordinate mapping system featuring a gesture-driven UI with sub-pixel accuracy.	
Disha — Blushlace & Dashboard <i>React, Node.js, MongoDB, Azure</i>	Code
- Produced comprehensive web platforms including a digital shelf-analytics dashboard that ingests anomaly data from Azure Blob Storage. - Established robust e-commerce and administrative architectures handling real-time KPI tracking and global state management.	
CheckIt (Quick Commerce Price Comparison) <i>React Native, Expo, Firebase</i>	Code

- Created a cross-platform mobile application to aggregate and compare prices in real-time across 7 major quick commerce platforms, helping users save on grocery bills.

Argo CD MCP Server | *TypeScript, Express.js, Model Context Protocol* Code

- Formulated an AI-powered Model Context Protocol (MCP) server for Argo CD, enabling AI assistants to interact with cloud-native applications via natural language.

EDUCATION

IIIT Bhubaneswar <i>B.Tech in Electronics & Telecommunication Engineering</i>	<i>2021 – 2025</i> <i>Bhubaneswar, India</i>
Central Academy <i>Senior Secondary (Class XII)</i>	<i>2018 – 2020</i> <i>Jodhpur, India</i>
Central Academy <i>Secondary School (Class X)</i>	<i>2016 – 2018</i> <i>Jodhpur, India</i>

ADDITIONAL

Interests: Playing badminton, Deep Learning, DSA, System Design, Web Technology.

Societies: Film & Theatrical Society (FATS) and Photogeeks Society, IIIT Bhubaneswar.